

Figure 28. Left $a + b$ heads generally occur late in the network, while a, b heads occur earlier. Mixed heads occur in middle layers. *Middle* $a + b$ heads attend more to the last token. *Right* a, b heads attend mostly to tokens a and b.

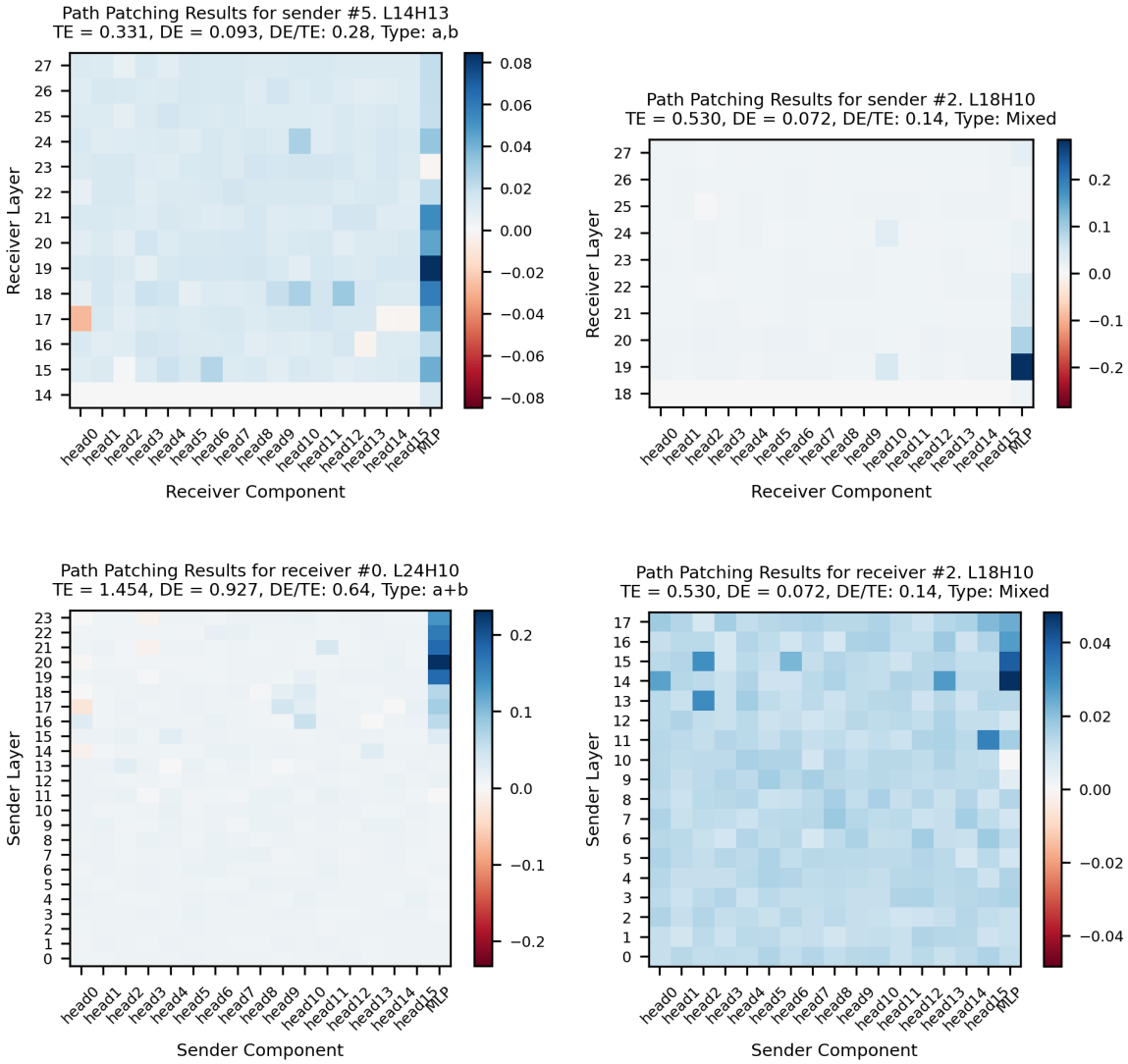


Figure 29. We present path patching results for top attention heads. In the top row, we view one a, b head (L14H13) and one mixed head (L18H10) as senders. We plot the total effect of each downstream component when only the path between the sender head and that component is patched into the model. We find that both a, b and mixed heads write primarily to MLPs. Similarly, when viewing an $a + b$ head (L24H10) and mixed head (L18H10) as receivers, we see that that the $a + b$ head is primarily dependent on the output from preceding MLPs. While that is true for the mixed head as well, we see that the mixed head also takes input from other a, b heads, implying that it could have some role in creating the $a + b$ helix.

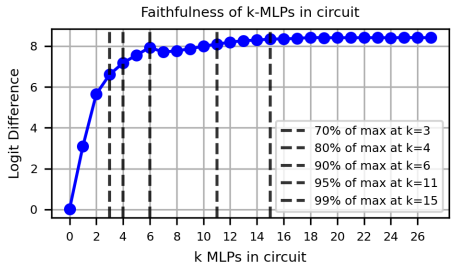


Figure 30. $k = 11$ MLPs are required to achieve 95% of the effect of patching in all MLPs.

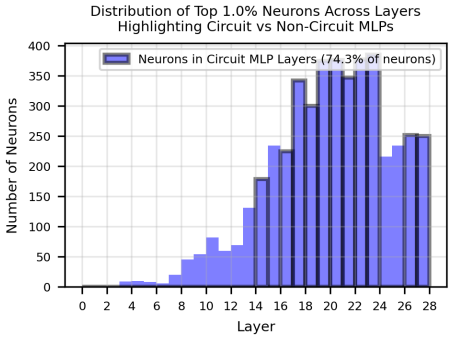


Figure 32. When we analyze approximately 1% of last token neurons most causally implicated in addition for GPT-J, we find that nearly 75% of them are in circuit MLPs, which is expected.

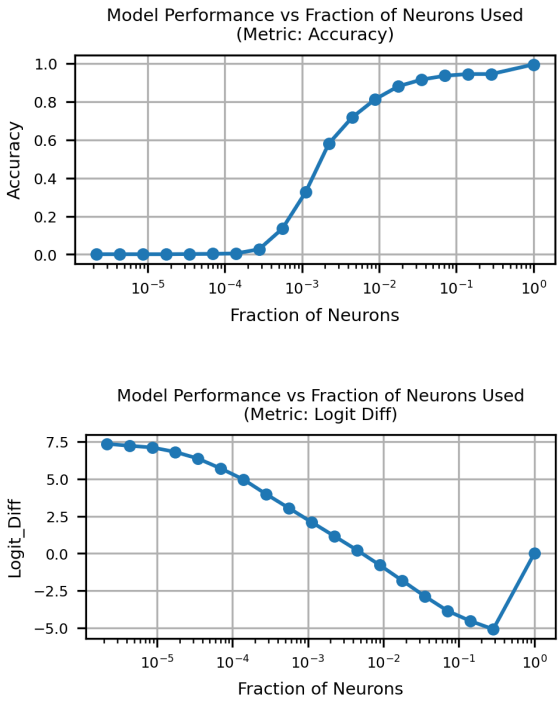


Figure 31. Top We see that using around 1% of top neurons and mean ablating the rest can restore the model to 80% accuracy. *Bottom* When measuring logit difference, we find that mean ablating some neurons on average *increases* the logit for the correct answer, but does not improve accuracy. We choose not to investigate this further.

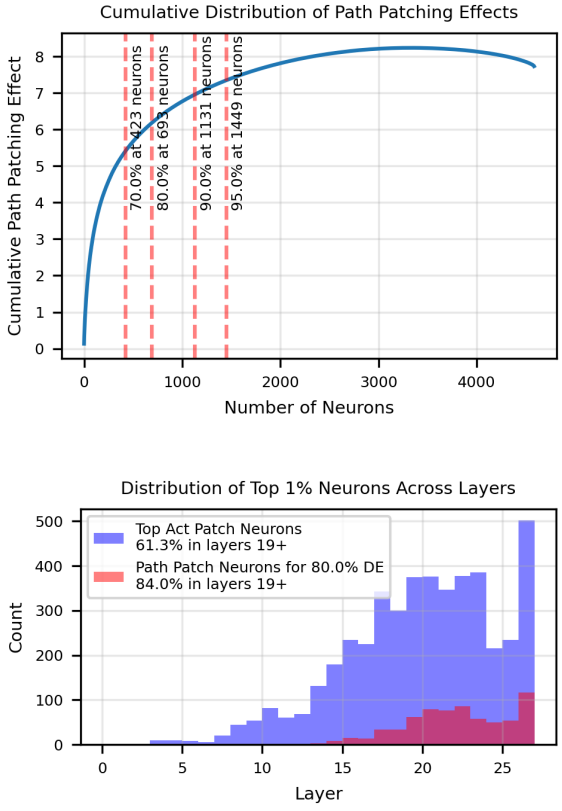


Figure 33. Top We find that roughly 700 neurons achieve 80% of the direct effect of patching in all $k = 4587$ high impact neurons. *Bottom* More than 80% of these high direct effect neurons are in layers 19 onwards, which we have identified as being responsible for translating the $a + b$ helix to answer logits.

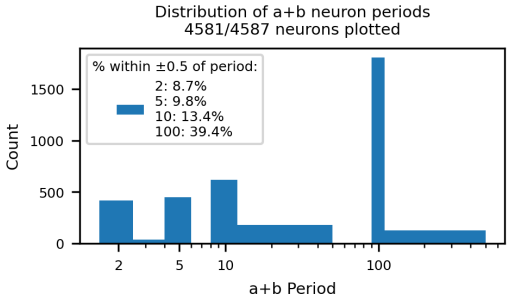


Figure 34. Top neurons' preactivations are periodic in $a + b$ with top periods of $T = [2, 5, 10, 100]$. Percentages shown for Fourier periods within 5% of T.

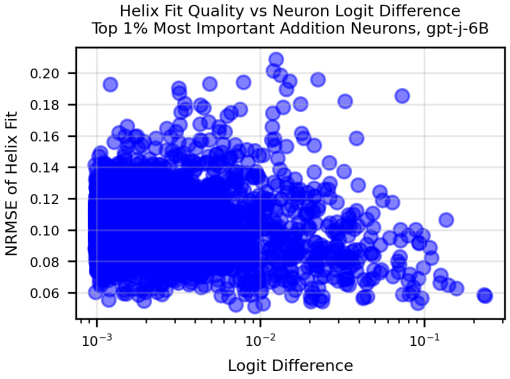


Figure 35. When calculating the NRMSE of the helix inspired fit for neuron preactivations, we find that more impactful neurons (with higher total effect) are typically fit better.

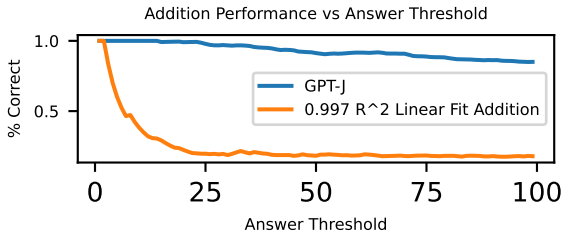


Figure 36. We find that using a line with $R^2 = 0.997$ leads to addition that performs considerably worse than GPT-J. We attribute this to the representational precision required to do addition along a line, indicating a possible reason LLMs choose to use helical representations.

has an R^2 of 0.997, indicating a very good fit. We consider all problems $a_1 + a_2$. We do addition on this line by taking $\ell(a_1) + \ell(a_2)$. If $\ell(a_1) + \ell(a_2)$ is closest to $\ell(a_1 + a_2)$, we consider the addition problem successful.

We then take the percentage of successful addition problems where the answer $a_1 + a_2$ is less than some threshold α, and compare the accuracy as a function of α for GPT-J and linear addition. Surprisingly, we find that for $\alpha = 100$, linear addition has an accuracy of less than 20%, while GPT-J has an accuracy of more than 80% (Fig. 36).

Thus, even with very precise linear representations, doing linear addition leads to errors. We interpret LLMs use of modular circles for addition as a built-in redundancy to avoid errors from their imperfect representations.

F. Investigating Model Errors

Given that GPT-J implements an algorithm to compute addition rather than relying on memorization, why does it still make mistakes? For problems where GPT-J answers incorrectly with a number, we see that it is most often off by -10 (45.7%) and 10 (27.9%), cumulatively making up over 70% of incorrect numeric answers (Fig 37). We offer two hypotheses for the source of these errors: 1) GPT-J is failing to "carry" correctly when creating the $a + b$ helix or 2) reading from the $a + b$ helix to answer logits is flawed.

We test the first hypothesis by analyzing the distribution of GPT-J errors. If carrying was the problem, we would expect that when the model is off by -10, the units digits of a and b add up to 10 or more. Using a Chi-squared test with a threshold of $\alpha = 0.05$, we see that the units digit of a and b summing to more than 10 is not more likely for when the model's error is -10 than otherwise (Fig. 38). This falsifies our first hypothesis. Thus, we turn to understanding how the $a + b$ helix is translated to model logits.

Since MLPs most contribute to direct effect, we begin inves-